DUALIZING COMPLEXES DESCEND ALONG A FAITHFULLY FLAT MAP OF NOETHERIAN RINGS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Descent of finiteness of modules along a faithfully flat map. Invoked inside the counted proof (source lines 4551-4554). Shows that the cohomology modules of the descended complex are finite over the source.
- The base change isomorphism for the derived internal hom along a flat map. Invoked inside the counted proof (source lines 5455-5459 and 5469-5470). Identifies the base change of the internal hom with the internal hom of the base changed complex.
- The criterion for finite injective dimension through the vanishing of the internal hom against finite modules. Invoked inside the counted proof (source lines 5465-5466). Converts the vanishing range of the internal hom into finite injective dimension of the complex.

Source reference. Original reference algebra-lemma-descend-properties-modules: sources/stacks-dualizing/pinned-original-source.tex, source lines 5454–5454. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-descend-properties-modules, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference more-algebra-lemma-base-change-RHom: sources/stacks-dualizing/pinned-original-source.tex, source lines 2719–2719. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-base-change-RHom, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's

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Source reference. Original reference more-algebra-lemma-injective-amplitude: sources/stacks-dualizing/pinned-original-source.tex, source lines 3299–3299. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-injective-amplitude, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $A \to B$ be a faithfully flat map of Noetherian rings. If $K \in D(A)$ and $K \otimes_A^{\mathbf{L}} B$ is a dualizing complex for B, then K is a dualizing complex for A.*

Proof. Since $A \to B$ is flat we have $H^i(K) \otimes_A B = H^i(K \otimes_A^{\mathbf{L}} B)$. Since $K \otimes_A^{\mathbf{L}} B$ is in $D^b_{Coh}(B)$ we first find that K is in $D^b(A)$ and then we see that $H^i(K)$ is a finite A-module by Algebra, Lemma algebra-lemma-descend-properties-modules. Let M be a finite A-module. Then

$$R\mathrm{Hom}_A(M, K) \otimes_A B = R\mathrm{Hom}_B(M \otimes_A B, K \otimes_A^{\mathbf{L}} B)$$

by More on Algebra, Lemma more-algebra-lemma-base-change-RHom. Since $K \otimes_A^{\mathbf{L}} B$ has finite injective dimension, say injective-amplitude in $[a, b]$, we see that the right hand side has vanishing cohomology in degrees $> b$. Since $A \to B$ is faithfully flat, we find that $R\mathrm{Hom}_A(M, K)$ has vanishing cohomology in degrees $> b$. Thus K has finite injective dimension by More on Algebra, Lemma more-algebra-lemma-injective-amplitude. To finish the proof we have to show that the map $A \to R\mathrm{Hom}_A(K, K)$ is an isomorphism. For this we again use More on Algebra, Lemma more-algebra-lemma-base-change-RHom and the fact that $B \to R\mathrm{Hom}_B(K \otimes_A^{\mathbf{L}} B, K \otimes_A^{\mathbf{L}} B)$ is an isomorphism. □